



- 2 A rowing club has a large number of members. A random sample of 12 of these members is taken and the pulse rate, x beats per minute (bpm), of each is measured after a 30-minute training session. A 98% confidence interval for the population mean pulse rate, μ bpm, is calculated from the sample as $64.22 < \mu < 68.66$.

(a) Find the values of $\sum x$ and $\sum x^2$. [6]

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(b) State an assumption that is necessary for the confidence interval to be valid. [1]

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